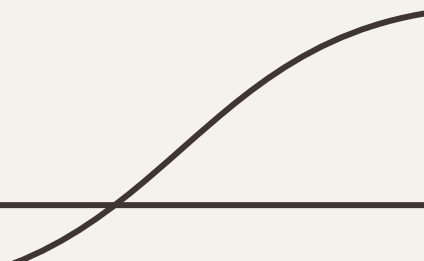


My Engineering Portfolio

Angela Suresh

Personal Website: angelasrsh.github.io



About Me

- **4th year ECE Student at University of Texas at Austin**
- Electrical Engineer on **Formula SAE EV Team for 3 years**
- Coursework focus: Antennas and RF, Digital Design, FPGAs, Embedded Systems, Signal Processing
- **Career goals: Hardware Engineer in Industry**
- Experience: **Two Summer Internships at Cisco, 3 years on FSAE EV team (in 4th year), Currently in an internship/Co-op at STMicroelectronics**
- Hobbyist Game Developer

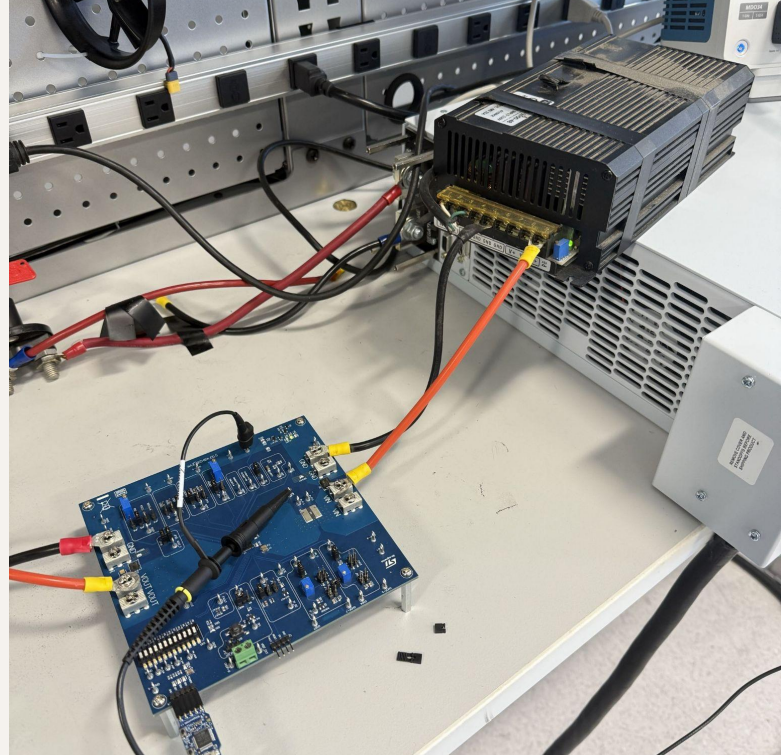


Internship: STMicroelectronics

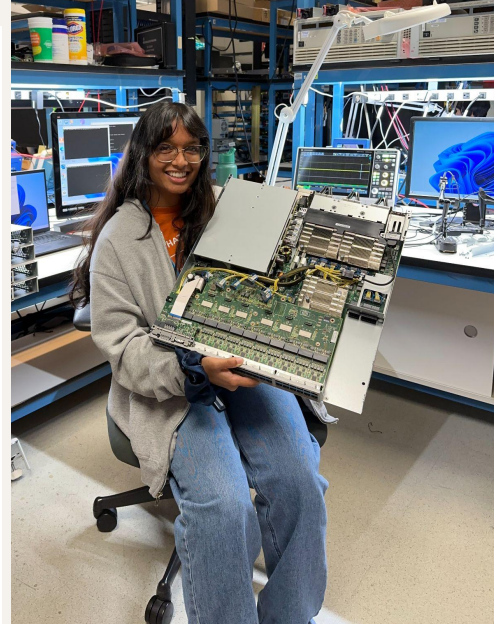
Role: PMIC Test Bench Validation Intern

Testing eFuse evaluation board using lab equipment.

Identified issues with test setup compatibility with the eval boards.



Internship Project: SFP28 Network Switch



Requirements

- SFP port compatible network switch with 25 Gb/s Uplinks
- Documented power budgets, schematic changes, test plan
- Hosted schematic reviews with experienced engineers
- Communicated with vendors for components
- Collaborated with various internal teams (ECAD, MCAD, Power, SI)



Previously at Cisco...

- Conducted hardware bringup for existing products
- Created and executed a detailed test plan for a rework
- Researched and simulated power converters and analyzed ripple effects



Formula SAE EV Team: Longhorn Racing

Yearly EV racing competition hosted in Michigan

1 year build cycle

Custom electronics with telemetry system and Grafana dashboard

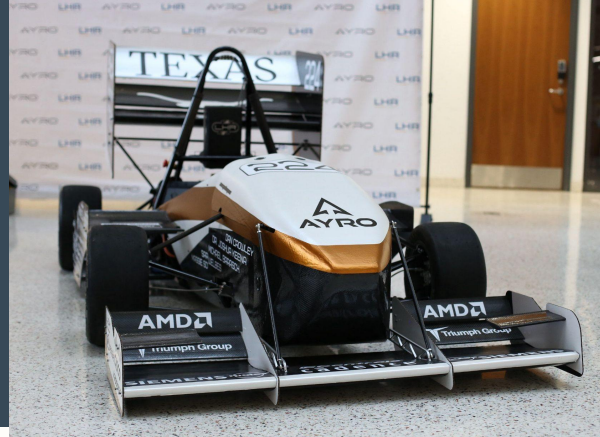
Board to board communication with CAN bus



Formula SAE EV Team: Longhorn Racing

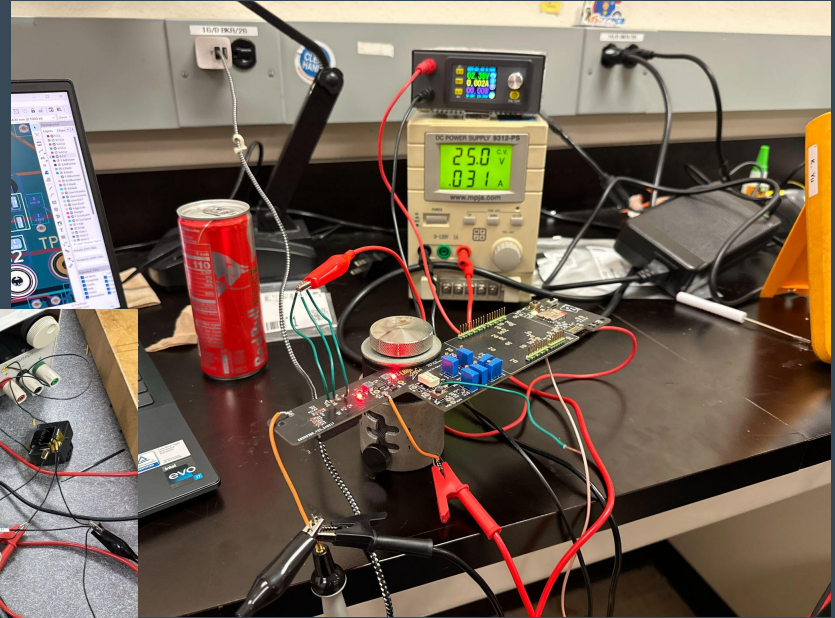
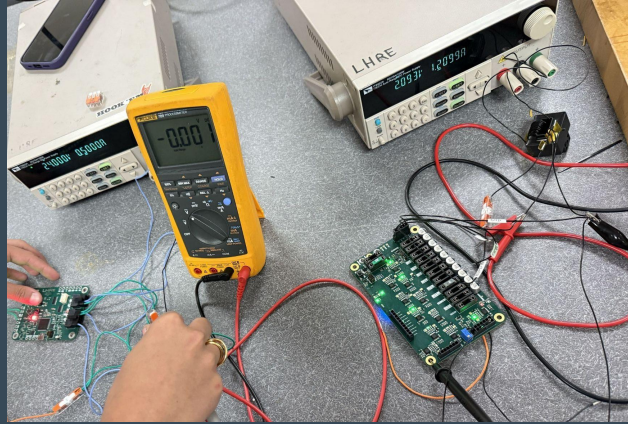
Skills I've acquired through the team:

- Circuit Design
- PCB Design
- Enclosure Design
- Firmware
- Hardware Bringup and Validation
- Requirements Documentation



Tools Used

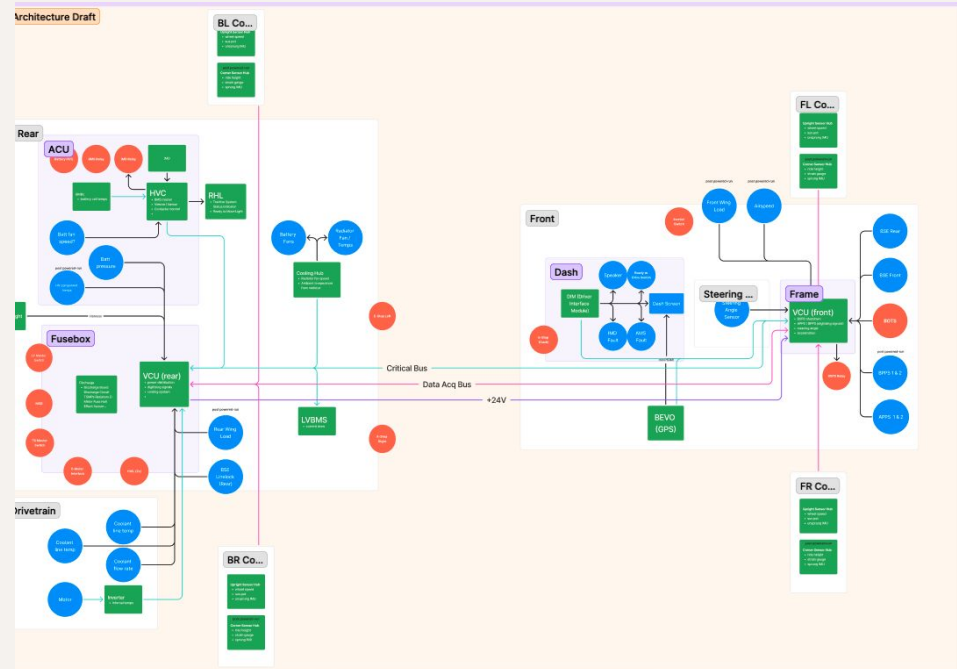
- Soldering
- Reflow Oven
- Logic Analyzer
- Power Supply
- DMMs
- Scopes



Doing hardware bringup for our Vehicle Control Unit
← And Power Distribution Unit

Other Work and Responsibilities

- Architecture design for LV system
- Knowledge transfers to younger members
- Reviewed schematic and layout designs
- Technical presentations over niche topics
- Recruitment and interviewing for new members

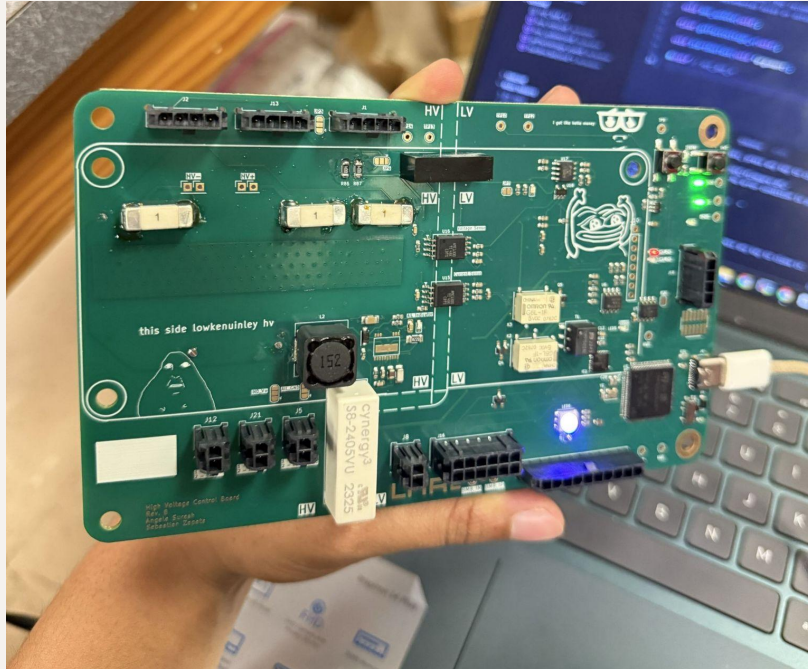


Lv architecture for 2025-2026 year- New 2 CAN bus topology

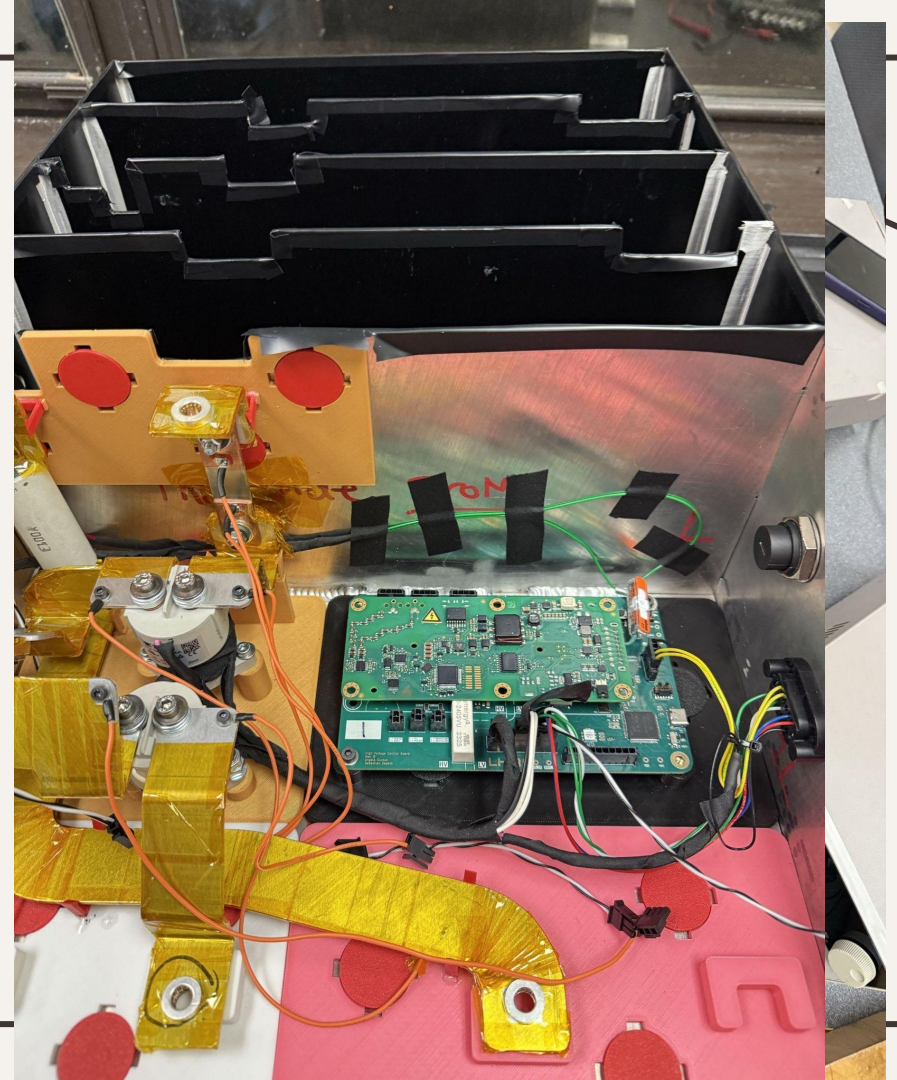
Current Project: Automotive Ethernet

- In progress project: to test viability and and a just for fun senior project
- Implementing 10BaseT 1S to reduce wire weight on the car, also used commonly in automotive industry
- **Requirements:**
 - 10BASE-T1S Compliance -multi drop bus mode that is similar to can- 10 Mbps
 - Capture sensor data via SPI output from MAC/PHY
 - Using STM32G4 MCU
 - Transmit analog and digital values via CAN
 - USB C
- **Tools:**
 - **Kicad** - schematic capture and PCB layout
 - **LTspice** - simulating Common Mode Choke effect with Ethernet TX/RX

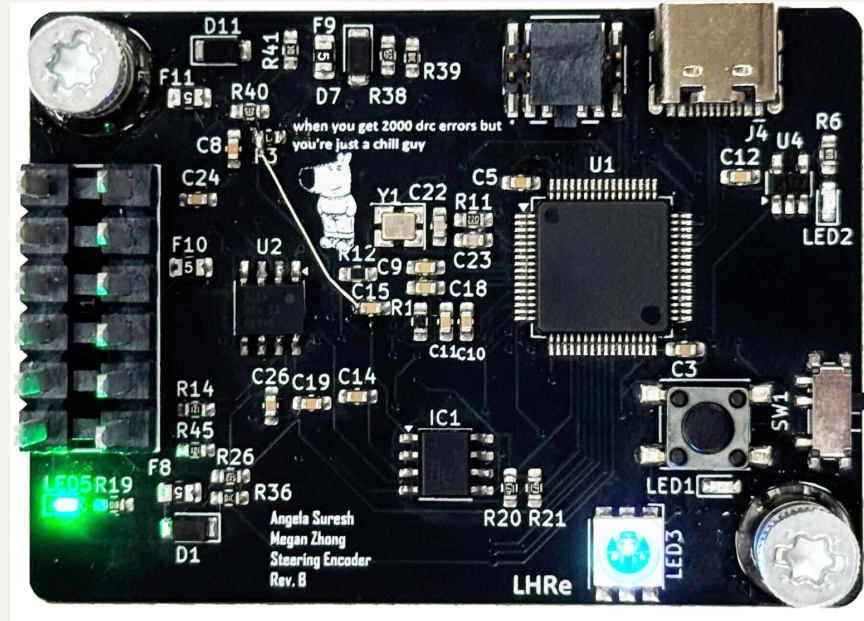
HW + Firmware



Senses voltage and current from battery pack, enables FSAE Rule regulated faults, and sends/receives data via CAN.



Steering Wheel Encoder



Design

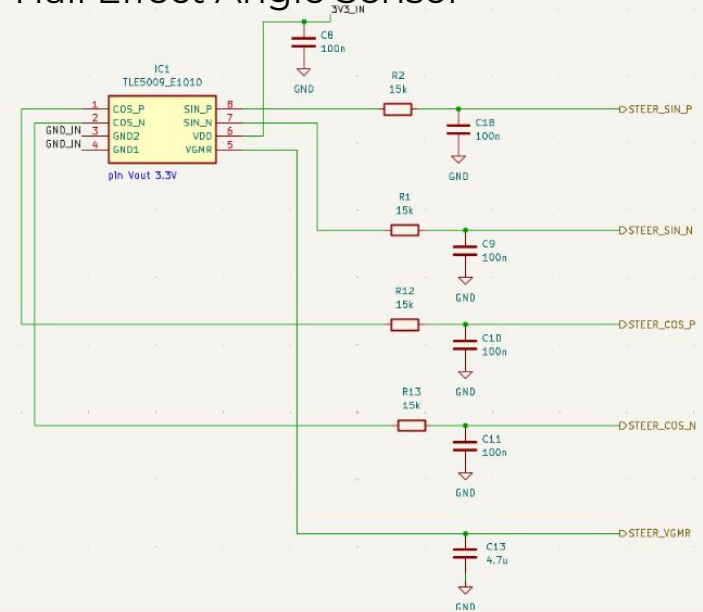
Requirements: Monitor steering wheel angle and send data via CAN to Vehicle Control Unit (separate board), monitor analog brake/accelerator pedal voltages, firmware written with C++

Process:

- Startup debugging
- Driver tuning for correct data acquisition
- Sensor calibration
- Board to board communication



Hall Effect Angle Sensor



Enclosure was not waterproof :(

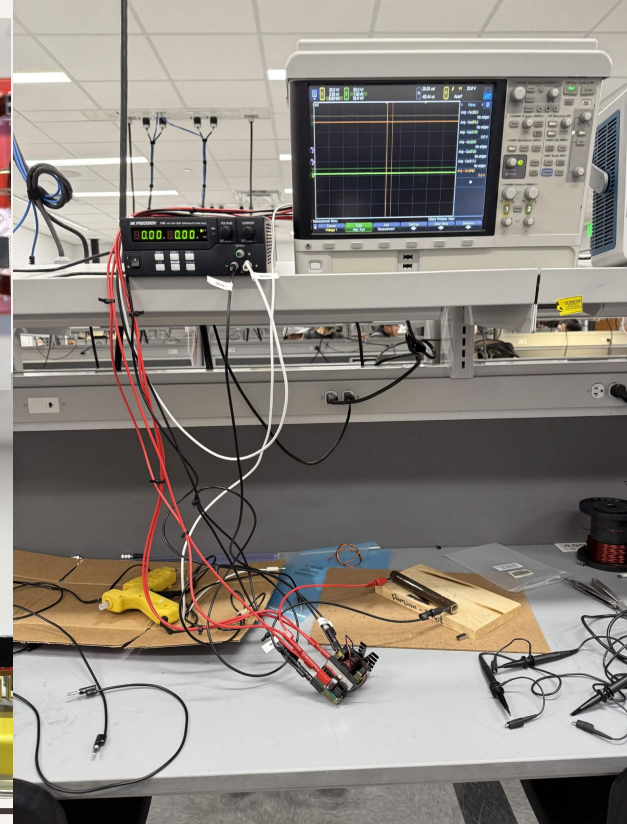
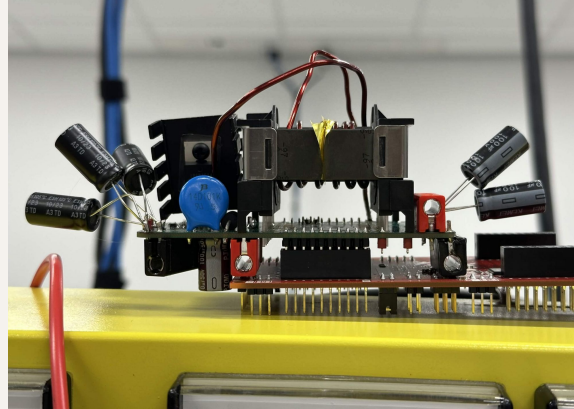
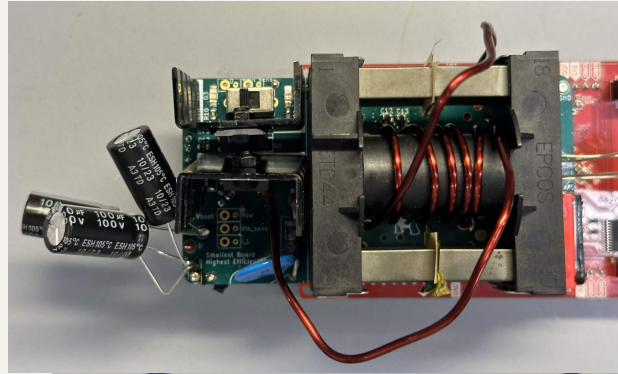
Boost Converter - Power Electronics

Requirements:

- Using TI C2000 MCU
- Custom Inductor Design
- 24 Vin to 48 Vout
- Operation in CCM
- 250W max Pout

Results:

- 96.8% efficiency (2nd in class)
- 2nd Place smallest Board



Circuit Protection Research

Requirements: Implement a low cost overvoltage and overcurrent protection system that can be universal to all boards on the car

- Simulated voltage and current inputs with benchtop supply
- Created test plan
- Selected low cost components and soldered necessary components when needed



Game Development - C#, Unity, Godot

Magic Capybara Rhythm Game

This mini game was created for the Egads mega micro 2022 game jam, made with Unity and C#. Check out the game integrated in the [mega micro game jam final game](#).

[More information](#) ^

Published  Mar 22, 2025

Status [Released](#)

Author [7Heaven](#)



John Chimp 3: Bullet Train

John Chimp 3: Bullet Train

Gameplay Instructions:

Point and CLICK with your mouse on the blue markers to choose where to go

Press SPACE to lock your movements

Press Q to cancel your movements

while moving through the markers you chose Point and Click with your mouse to shoot your gun

Use ONLY arrow keys to move around

Click on green arrows to progress

DONT PRESS THE LETTER D, YOU WILL DIE

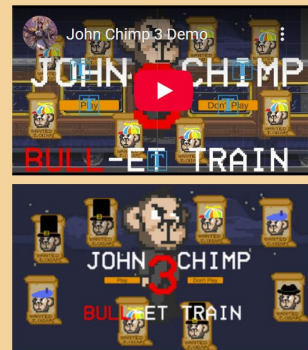
THE TEAM:

GAME Design + MECHANICS: Arnold, Luke, Eric, Angela

Music + SFX: Luke, Arnold

Art: Angela, Arnold

Story: Eric



<https://angelasrsh.itch.io/>

Questions?

